

ASSIGNMENT 4.2

Q1. List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).

NO.	Hardware Issue	Typical Symptom (what you see/hear)
1	Power Supply Unit (PSU) failure	PC does not turn on at all; no fan spin, no LED, no POST beep.
2	RAM (memory) fault	PC powers on but no display; sometimes repeated beep codes or random blue screens/freezes.
3	Overheating CPU / failed CPU fan	System shuts down suddenly under load; BIOS may show CPU temperature warning; fan noise abnormal or silent.
4	Hard Drive / SSD failure	Very slow boot, clicking/grinding noise (HDD), "No boot device" or "Disk read error" messages.
5	Graphics card (GPU) failure	No display even though PC seems on; artifacts on screen before crash; driver crash messages in OS.

Q2. Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

Step 1 - Check the Wall Outlet

- - Plug another device (lamp/phone charger) into the same outlet.
- - If other device also dead: outlet/switch/fuse is the issue. Try a different outlet.
- - If other device works: outlet is OK; move to next step.

Step 2 - Inspect Charger Cable and Brick

- - Visually check entire cable for cuts, kinks, exposed wires.
- - Check the brick for burn smell, bulging, or loose parts.
- - Ensure both ends (wall side and laptop side) are firmly plugged in.

Step 3 - Check the Laptop Charging Port

- - With laptop OFF and unplugged, look into the DC-in port for dust, bent pins, or looseness.
- - Gently wiggle the connector while plugged in (laptop off).
- - If LED on brick or laptop charging indicator flickers with wiggle: port may be loose/damaged.

Step 4 - Power Reset (Drain Residual Charge)

- - Shut down laptop, unplug charger, remove battery (if removable).
- - Hold power button for 15-30 seconds.
- - Reconnect battery (if removed), plug in charger, try powering on.

Step 5 - Test Charger Output with Multimeter (if available)

- - Set multimeter to DC Voltage (20V range).
- - Identify charger DC plug polarity (usually center-positive; printed on brick).
- - Touch red probe to inner pin, black probe to outer sleeve.
- - Expected: Voltage close to rated output (e.g., 19V plus/minus ~1V depending on adapter).
- - If 0V or far below rating: charger is faulty -> replace.

Step 6 - Try a Known-Good Compatible Charger

- - If possible, test with another charger of same voltage and equal/higher wattage and correct connector.
- - If laptop charges with other adapter: original charger is bad.
- - If still not charging: suspect laptop charging port, internal power circuit, or battery.

Q3. Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

Setup (real or simulator):

- - Set multimeter to DC Voltage, 20V range.
- - Identify USB-A pins: VCC (+5V) = outer right pin (when metal side facing you, pins up); GND = outer left pin.
- - Touch red probe to VCC pin, black probe to GND pin.

Typical Reading & Interpretation:

Measured Voltage (example): 5.02 V

Expected USB range: 4.75 V - 5.25 V

Conclusion: 5.02 V is within 4.75-5.25 V -> charger output is normal and safe for USB devices.

4) A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate? Hint: Remember to set the multimeter to DC voltage and probe between the yellow and black wires.

Steps to Measure 12V Rail (Yellow Wire)

- - Power off and unplug PC.
- - Disconnect 24-pin ATX connector from motherboard.
- - Set multimeter to DC Voltage, 20V range.
- - Identify wires on ATX connector: Yellow = +12V rail; Black = Ground (0V).
- - Insert red probe into yellow wire pin, black probe into any black wire pin.
- - Turn on PSU (using paperclip method if required).
- - Read voltage on multimeter.

Example Reading & Interpretation:

Measured Voltage (example): 11.8 V

Acceptable 12V rail range: roughly 11.4 V - 12.6 V (plus/minus 5%). Many guides flag anything below 11.5 V as suspect.

- **If reading is below 11.5 V (e.g., 11.2 V):**
 - - This indicates the 12V rail is undervoltage, which can cause random shutdowns under load, instability in CPU/GPU and drives, and possible failing or overloaded Power Supply Unit (PSU).
 - - Likely hardware issue: Weak/failing PSU or PSU unable to handle system power draw -> recommend testing with a known-good PSU or replacing the PSU.